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Psychological Sources of Context Effects in Survey Measurement

Survey research methodologists have long been aware and concerned that survey responses are context-dependent. Most important, early questions may influence responses to subsequent ones (see Bradburn, 1983; Payne, 1951; Schuman & Presser, 1981; Schwarz & Strack, 1991a; Schwarz & Sudman, 1992; Tourangeau & Rasinski, 1988, for research examples and reviews). However, the conditions under which context effects occur are not well understood, and when they do occur, it is difficult to predict their direction.

In this and the following chapter, we explore the different processes that underlie the emergence of context effects. In the present chapter, we discuss their various psychological sources and review selected findings from the survey methodology literature. In Chapter Five we focus on variables that determine whether responses to subsequent questions become similar to or different from responses to preceding questions, exploring the conditions that elicit assimilation or contrast effects in attitude measurement. The final section of Chapter Five links our observations on context effects to traditional theorizing in the domain of attitude research, asking what the implications of context effects are for how we think about the nature of attitudes in general. Finally, in Chapter Six we address a specific issue in attitude measurement, namely, the impact of the order in which response alternatives are presented.

Context Effects

Before exploring the nature of context effects, it is important to emphasize that researchers always set contexts that influence some aspect of the question answering process. Sometimes we do so deliberately—for example, to clarify the meaning of a question or encourage respondents to consider an issue in the context that is of interest to the researcher—but often we do so unintentionally. What has been called “response effects” or “response errors” in the survey methodology literature may be thought of as those cases where we unthinkingly affect the context and thus, for the purpose at hand, create “error.”

However, the concept of “error” is quite problematical in attitude measurement. Whereas reports about behaviors or events can, at least in principle, be verified, attitude reports reflect subjective evaluative judgments. Yet human judgment is *always* context-dependent; in essence, context-free judgments do not exist. Accordingly, if we want to talk of “errors” in attitude measurement, we can only do so in relation to what we are trying to do in a question-naïve: there is no objective standard that reflects respondents’ “true” attitudes. Thus, to avoid the “error” terminology when we address attitude measurement, we speak of context effects instead.

Many context effects are the result of the influence of question order. They have often been referred to as question order effects in the survey methodology literature. However, from a theoretical point of view, question order is simply a technical aspect of the questionnaire and does not by itself constitute a meaningful psychological variable. Rather, the content of a preceding question may affect any of the steps of the question answering process (outlined in Chapter Three) and involve a number of different psychological processes. Moreover, any given order effect observed in a survey may reflect the impact of several of these processes. Disentangling the relative contribution of different processes in a given case presents a difficult task, and some of the more robust “classic” context effects are likely to be overdetermined. Furthermore, we

can only speak of order effects in the strict meaning of the term when the questions are presented in a face-to-face or telephone interview. In mail surveys and other self-administered questionnaires, respondents may read later questions before answering earlier ones or backtrack and change previous responses, thus obliterating the presentation order. In such cases, later questions may influence responses to earlier ones. Although systematic information on how often respondents read ahead or return to previous questions is not available, several studies found an impact of subsequent questions on the answers given to previous ones (for example, Bishop, Hippler, Schwarz, & Strack, 1988; Schwarz & Hippler, 1995), as we shall see in some of the examples reviewed in the following paragraphs. Before addressing any examples in detail, however, let us look at the cognitive processes that are likely to be involved in the emergence of context effects.

First, the content of preceding questions may influence respondents' interpretation of later questions (or subsequent ones may influence interpretation of earlier ones, in the cases noted earlier), resulting in context effects at the comprehension stage, as we discussed in Chapter Three.

Second, answering a preceding question may render the information used to answer it more accessible in memory and thereby increase the likelihood that it be used in forming subsequent judgments to which it may be relevant. As a result we may obtain context effects at the judgment stage. In most cases, the information rendered more accessible in memory will be substantive information, typically some feature of the target of judgment. For example, answering a question about the president's performance on economic issues increases the likelihood that this aspect of his performance will be considered in evaluations of his overall performance. In other cases, the information rendered accessible may be a general norm, which may subsequently be applied to other issues. Both cases have been given considerable attention in survey methods research.

However, our judgments do not depend only on *what* comes to mind. Consideration of an issue may include a variety of phenomenal experiences, such as our affective reaction to the content we are thinking about or the ease with which that content comes to mind. Such phenomenal experiences constitute a source of information in their own right, and different preceding questions may elicit different phenomenal experiences. To date, this source of context effects has received only limited attention in survey research.

Answering a question may also increase the accessibility of cognitive procedures, thereby increasing the likelihood that the same procedure will be employed later on. Whereas the activation of substantive or normative information and differences in the phenomenal experiences that accompany the thought process all influence *which* information respondents use in computing a judgment, the activation of procedural knowledge influences *how* respondents use the information, that is, which strategy they employ.

In addition, preceding questions may influence what comes to mind when respondents anchor a rating scale, thus influencing the response formatting process. Finally, preceding questions may increase or decrease respondents' concerns about self-presentation and social desirability, affecting the editing stage of the question answering sequence. In the following sections, we explore each of these processes in turn.

Context Effects at the Comprehension Stage

We have already addressed context effects at the comprehension stage in Chapter Three, noting that they may reflect the operation of semantic priming processes as well as the operation of pragmatic inferences (see Strack, 1992, for a detailed discussion). If respondents encounter an ambiguous question, they are likely to interpret it in terms of the applicable concept that is most accessible at that point in time (see Higgins, 1989; Wyer & Srull, 1989, for reviews). Moreover, they may deliberately refer to preceding questions to

infer the intended meaning of the ambiguous one, as we have seen in the Strack, Schwarz, and Wänke (1991) study reviewed in Chapter Three. In that study, German students interpreted a question about a fictitious "educational contribution" to mean that students would have to pay money when the question was preceded by another about tuition fees and to mean that students would receive money when it was preceded by a question about student support payments. (See also Tourangeau & Rasinski's [1988] discussion of an obscure monetary control bill, first introduced by Schuman & Presser, 1981.)

In general, the more ambiguous the question the more likely context effects are to emerge at the comprehension stage. It is under this condition that respondents are most likely to turn to the context to make sense of a question (see Schwarz, 1994; Strack & Schwarz, 1992, for reviews). However, there is an important exception to this generalization. As we discussed in Chapter Three, according to the rules that govern the conduct of conversation in everyday life speakers must provide information that is new to the recipient rather than reiterate information that the recipient already has (see Clark, 1985). Application of this rule to the survey interview may result in different interpretations of questions that are not in themselves particularly ambiguous. For example, Schwarz, Strack, and Mai (1991; see also Tourangeau, Rasinski, & Bradburn, 1991) first asked respondents to report their happiness with their marriage and subsequently their satisfaction with their lives as a whole. When these questions were presented on different pages of the questionnaire, respondents interpreted the question about their life as a whole to include their marriage, much as one would expect. For other respondents, however, both questions were placed in the same conversational context by a joint lead-in that read, "We now have two questions about your life." In this case, respondents interpreted the general life-satisfaction question as if it were worded, "Aside from your marriage, which you already told us about, how satisfied are you with *other* aspects of your life?" We will return to this study in more detail in the next chapter. For now we

note that preceding questions may change the interpretation of subsequent ones even under conditions where the wording of the question is not particularly ambiguous. This is to be expected when the two questions form a part-whole relationship, such that a specific preceding question renders a subsequent more general question partially redundant. In that case, the general question may be interpreted as excluding the specific aspect that has already been assessed.

Context Effects at the Judgment Stage

Most context effects at the judgment stage reflect the increased accessibility of information that was used to answer preceding questions. They may take different forms depending on whether the information rendered accessible is a previously formed judgment, a feature of the target of judgment, or a general norm. In addition, the process of answering a question may elicit subjective experiences, such as whether the question is easy or difficult to answer, and these experiences may themselves provide information that is used in making subsequent judgments. We elaborate on these processes in the following section, and we return to some of them in more detail in the next chapter.

Previously Formed Judgments

In some cases, a question may prompt respondents to form a judgment that may bear on a subsequent related question. If the previously formed judgment is still accessible in memory, respondents may recall this judgment and use it to derive an answer to the second question without retrieving much additional information. This possibility is nicely illustrated in a study by Carlston (1980). He asked subjects to form an impression of a student who allowed a fellow student to cheat on an exam by copying his answers. Some subjects were first asked to evaluate if the target person was "honest," whereas others were asked if he was "kind." The answer is obviously

"no" to the honesty question but "yes" to the kindness question. Yet when subjects who first evaluated the target's honesty were later asked to rate his kindness, they judged him to be less kind than did subjects who evaluated his kindness first. Conversely, those who had first evaluated his kindness went on to rate him as more honest than did those who evaluated his honesty first. This pattern of findings suggests that subjects did not recall the behavioral information they had used to form the first judgment when they were asked to make the second judgment. Rather, they seemed to use the evaluative implications of the first judgment to derive the second one, assuming that "good (or bad) traits go together." Therefore, if he is kind he must be honest, and vice versa (for additional examples, see Schwarz & Bless, 1992b).

In general, respondents who have formed a judgment are unlikely to start from scratch when asked to make a second, closely related judgment; rather they will probably derive the second judgment from the implications of the first, without reconsidering the original information used to form the initial judgment.

Features of the Target of Judgment

When respondents cannot recall a previously formed related judgment to derive a subsequent one, they need to start from scratch by recalling relevant information. The information most likely to come to mind is the information they used in answering related preceding questions. Accordingly, preceding questions may influence the information respondents use in constructing a mental representation of the target. We address the complexities of this process in greater detail in the next chapter, where we discuss how it determines the emergence of assimilation and contrast effects.

General Norms

A context effect that reflects the activation of a general norm was first reported by Hyman and Sheatsley (1950). In 1948, they asked

Americans if "the United States government should let Communist reporters from other countries come in here and send back to their papers the news as they see it." When this question appeared in the first position, only 36 percent of the American sample supported freedom of press for such reporters. However, when respondents were first asked if "a Communist country like Russia should let American newspaper reporters come in," a proposition that most respondents endorsed, support for Communist reporters in the United States increased to 73 percent. Endorsing freedom of the press for American reporters in Russia apparently made respondents aware that the same principle should hold for Communist reporters in the United States.

In subsequent research, Schuman and colleagues (see Schuman & Ludwig, 1983; Schuman & Presser, 1981) traced this and related findings to the operation of a norm of "reciprocity" or "evenhandedness." Although the empirical evidence is currently limited to the impact of reciprocity norms, we may expect that any question or sequence of questions that draws respondents' attention to a general norm is likely to elicit an application of the norm in making subsequent judgments to which the norm is applicable.

Subjective Experiences

The processes considered so far have focused on the *content* that is brought to mind when one answers a question. Sometimes, however, answering a question may elicit a variety of feelings. For example, thinking about a happy or sad event may induce a happy or sad mood (for example, Strack, Schwarz, & Gschneidinger, 1985). If so, the mood may itself influence subsequent judgments, even about targets that are unrelated to the specific content that elicited the mood in the first place. In a large number of studies, being in an elated or depressed mood has been shown to influence a wide range of judgments, from satisfaction with consumer goods (for example, Isen, Shalcker, Clark, & Karp, 1978) to feelings toward other people or estimates of risk (for example, Johnson & Tversky, 1983). In

general, respondents provide more positive evaluations when they are in a good mood (see Forgas, 1992; Schwarz, 1990b; Schwarz & Clore, 1988, for reviews). In such cases respondents may simplify the judgmental task by using their affective reaction to the target as a basis of judgment. Asked, for example, if one likes something, one may consult one's feelings about the target, essentially asking oneself, "How do I feel about it?" In so doing, however, it is difficult to distinguish between one's affective reaction to the target and one's preexisting mood. As a result, the target is evaluated more positively when people are in good moods than when they are in bad moods. Such mood effects are not obtained, however, when people are aware that their feelings may not reflect their reaction to the target but may be the effect of something else.

A study by Schwarz and Clore (1983, Experiment 2) serves as an illustration. Respondents who were interviewed by telephone reported higher general life satisfaction when they were called on sunny days than when they were called on rainy days. Asking themselves, "How do I feel about my life as a whole?" they evaluated their life more positively when they felt good rather than bad. The impact of the weather, however, was eliminated when the interviewer opened the conversation with a private aside, asking the respondent, "How is the weather down there?" Drawing attention to the weather made respondents aware that their current mood might not reflect how their lives were going in general but only the impact of a transient cause, namely, the weather. Thus, the informational value of their current mood was called into question and its impact was eliminated (see Schwarz & Strack, 1991b, for a more detailed discussion).

In general, questions that change respondents' current mood by bringing positive or negative material to mind are likely to affect subsequent judgments even if the target of judgment is completely unrelated. However, the less respondents are aware that their current feelings may have been caused by the question asked the more their mood is likely to have an impact on unrelated judgments. If the question is blatant, respondents may become aware that think-

ing about the question has affected their feelings and may disregard such feelings when asked to make another judgment, much as they did in the weather study. But if the affect-eliciting question is more subtle, respondents may see no reason to discount their feelings, resulting in mood-congruent judgments on subsequent questions (see Schwarz & Clore, 1988). Similar considerations hold when respondents are aware that a certain thought may come to mind because of a preceding question, as we will see in Chapter Five.

Although the role of subjective experiences in judgment has been most extensively studied in the domain of moods and emotions, other experiences may affect judgments in a similar way (see Banaji, Blair, & Schwarz, 1995; Clore, 1992). For example, respondents may find it easy or difficult to answer a certain question. This subjective experience alone may provide relevant information for making certain judgments. This assumption is at the heart of Tversky and Kahneman's (1973) availability heuristic, which holds that people estimate the frequency, likelihood, or typicality of events by the ease with which they can bring relevant examples to mind. Moreover, the ease or difficulty of recall may change the conclusions that respondents draw from the content they recall. A study by Schwarz, Bless, Strack, Klumpp, Rittenauer-Schatka, and Simons (1991) illustrates this possibility. They asked respondents to describe either six or twelve situations in which they behaved assertively. Whereas describing six examples was easy, it was difficult for respondents to call twelve examples to mind, although all could complete this task. Subsequently, they had to rate their own assertiveness. Theoretically, if the respondents relied solely on the examples they brought to mind they should have rated themselves as more assertive the greater the number of examples. However, this was not the case. Respondents rated themselves as more assertive when they had to recall only six examples. Apparently, the difficulty of recalling twelve examples suggested to them that they couldn't be all that assertive if it was so difficult to bring twelve examples to mind. Thus, the conclusions that respondents drew from the recalled examples depended primarily on whether the

recall task was experienced as easy or difficult. In a similar vein, Bishop (1987) observed that respondents were less likely to report that they followed public affairs if the question was preceded by a difficult knowledge question. Specifically, the knowledge question asked that they report what their United States representative had done for the district. Knowing little about the representative's record, respondents apparently concluded that they didn't follow what was going on.

It should be noted, however, that these findings should only be obtained when respondents attribute the experienced difficulty to their own inadequacy. These findings should not be obtained when they attribute the difficulty to another cause. In such cases, the experienced difficulty is not informative, much as we have seen for one's mood in Schwarz and Clore's (1983) weather study. In line with this assumption, Schwarz, Bless, et al. (1991) found that respondents reported higher assertiveness after recalling twelve rather than six examples when they were able to attribute the experienced difficulty to the impact of a distracting task. Similarly, Schwarz and Schuman (1992) observed in an extended replication of Bishop's (1987) studies that asking respondents to evaluate the public relations work of their U.S. representative eliminated the otherwise obtained negative effect of the knowledge question on their reported political interest, as shown in Table 4.1.

Confirming Bishop's (1987) results, Schwarz and Schuman (1992) found that asking a knowledge question about the performance record of the respondents' U.S. representatives increased the percentage of respondents who reported that they follow politics only "now and then" or "hardly ever," from 21 percent to more than 39 percent. Respondents apparently concluded from the difficulty they experienced in answering the knowledge question that they didn't follow politics very closely. In a third condition, however, respondents had to evaluate the quality of their representative's public relations work before they reported on how closely they follow politics. In this case, respondents could conclude that their lack of knowledge reflected the representative's poor job of keeping the

TABLE 4.1 Impact of a knowledge question on reported political interest.

	Question order		
	Political interest Knowledge	Knowledge Political interest	Knowledge Public relations Political interest
	21.0%	39.4%	29.6%

Source: Schwarz and Schuman, 1992.

Note: Shown is the percentage of respondents who reported following politics "now and then" or "hardly ever."

public informed rather than their own lack of interest. As a result, the impact of the knowledge question was cut by half, and only about 30 percent of the respondents reported that they followed politics only "now and then" or "hardly ever."

The latter finding is particularly interesting because Bishop (1987) observed that the impact of the knowledge question on reported political interest was largely unaffected by filler items, even under conditions where the two questions were separated by a full 101 unrelated buffer items. In contrast, a single intervening question was sufficient to reduce the effect to nonsignificance in the present study. This one question, however, was targeted to undermine the psychological process that drives the observed context effect. In combination, the differential impact of 101 versus 1 buffer items demonstrates that researchers need to understand the psychological processes that underlie a given context effect. Purely technical analyses, focusing on the sheer number of buffer items, for example, are unlikely to advance an understanding of survey response.

As these examples show, early questions may affect subsequent judgments by influencing not only what comes to mind but also the manner by which things come to mind. Feelings—that it is easy or difficult to answer a question, or that one is in a good mood—may themselves influence the conclusions drawn. However, respondents

will only use these experiences as a basis for judgment if they seem relevant to the judgment at hand. If they attribute the experienced feelings to a source that renders them irrelevant to the judgment at hand, they will disregard them. (See Clore, 1992; Schwarz, 1990b, for more detailed discussions of the informational value of feelings and subjective experiences.)

Context Effects at the Formatting Stage

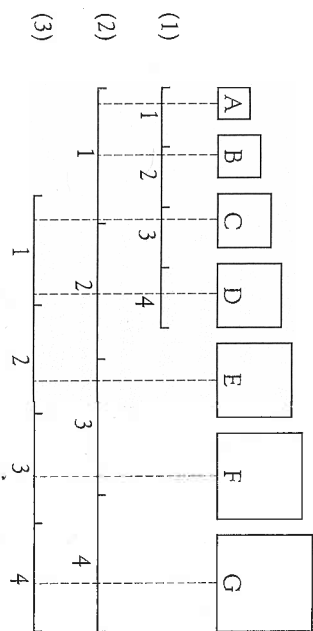
Preceding questions may also influence how respondents use the response scales provided to them, resulting in context effects at the formatting stage. This is particularly likely to happen when the response has to be provided along a rating scale, as described in the following section. In addition, the order in which the response alternatives of a categorical scale are presented may influence the likelihood that a given alternative is endorsed. However, this reflects the impact of the ordering of the response alternatives themselves rather than the impact of preceding questions. Accordingly, we address response order effects on categorical scales separately in Chapter Six.

Rating Scales

Numerous studies in psychophysics demonstrate that ratings of a stimulus are context-dependent (see Poulton, 1989, for a review). For example, a given square is rated as larger when presented in the context of smaller squares than when presented in the context of larger squares. This suggests that respondents anchor the rating scale on the extremes of the stimulus continuum, as shown in Figure 4.1.

Suppose that respondents were asked to rate the size of square C along a four-point scale, ranging from one (very small) to four (very large). If only squares A to D were presented to them, thus constituting the relevant "perspective" for this task (Ostrom & Upshaw, 1968), they would use A and D to anchor the scale. In this case, square C would receive a rating of three. Suppose, however,

FIGURE 4.1 Anchoring a rating scale.



that squares A to G were presented, extending the relevant perspective. In this case, anchoring the rating scale on A and G would essentially stretch the scale and square C would now receive a rating of two. Moreover, more than one square would be assigned to the same rating scale category, decreasing the discrimination between stimuli. Finally, suppose that only squares C to G were presented to respondents. In this case, anchoring the scale on C and G would result in a rating of one for square C, again reflecting a change in perspective.

Perspective effects of this type are most pronounced when all stimuli are rated along the same scale, as in this example. However, respondents can only anchor the scale on the extremes of the presented stimulus continuum when they are aware of what the extremes are. If only one stimulus is presented at a time, respondents will anchor the scale on extremes recalled from memory, reflecting whatever perspective is accessible to them at the time of judgment. To enter into the perspective, the extreme stimulus must seem relevant to the task at hand. Two studies from different domains illustrate this point.

In a classic study in psychophysics, Brown (1953) asked subjects to rate the weight of different blocks they were asked to lift. Not surprisingly, a given block was rated as heavier when preceded by a light block than when preceded by a heavy block. However, this was only the case when the heavy block was perceived as being part of the series of blocks to be rated. When the subjects were asked to

lift a tray—a dissimilar object—of equivalent weight, their judgment of the target objects remained unaffected. Apparently, the tray was not perceived as part of the stimulus series. Accordingly, it did not enter the subjects' representation of the relevant "perspective," and was ignored when anchoring the rating scale. As this example illustrates, even psychophysical judgments, which are often thought to depend primarily on sensory inputs, are strongly affected by the cognitive representation of the stimuli.

Although the Brown (1953) study demonstrates that an extreme stimulus must be perceived as relevant to the set of target stimuli in order to affect their ratings, it is not necessary that the extreme stimulus itself be rated. For example, Schwarz, Münkkel, and Hippler (1990; see also Noelle-Neumann, 1970) asked German respondents to rate how "typically German" a variety of drinks were. Before this task, some respondents had to estimate how many Germans drink beer (a typical drink in that country), whereas others had to estimate how many Germans drink vodka (an atypical drink). Their subsequent ratings of wine, milk, and coffee showed the expected contrast effect, as seen in the first row of Table 4.2.

Respondents who had estimated the consumption of beer rated all beverages as "less typically German" than respondents who had estimated the consumption of vodka, indicating that all used the highly accessible extreme stimuli brought to mind by the preceding

TABLE 4.2 Impact of extreme stimuli on subsequent ratings.

	Context stimulus	
	Vodka	Beer
Preceding question		
Consumption	5.4	4.4
Calorie content	4.4	4.5

Source: Adapted from Schwarz, Münkkel, and Hippler, 1990.

Note: $N = 25$ to 27 per cell; 9 = "very typical." Shown is the mean of ratings of three beverages (milk, wine, and coffee).

question to anchor the rating scale. It should be noted that this shift in ratings occurred even though the extreme stimuli themselves were not rated along the scale; this suggests that their sheer accessibility in memory was sufficient to change respondents' perspective. Reiterating the theme of Brown's (1953) study, however, this conclusion comes with one important qualification. In this example, respondents had to estimate the frequency with which Germans consume beer or vodka; this judgment was relevant to the typicality dimension because a drink is only "typically German" if Germans drink it frequently. But other respondents had to estimate the caloric content of a glass of beer or vodka. Whereas this task also served to increase the accessibility of the extreme stimuli in memory, it did not bear on the typicality issue. As a result, these respondents did not show a shift in their subsequent ratings, as seen in the second row of Table 4.2. Hence, we conclude that extreme stimuli brought to mind by preceding questions will only be used in anchoring the response scale if the content of the preceding question links them to the underlying dimension of judgment. If it does, these stimuli are likely to affect subsequent ratings even if they themselves are not rated along the same scale. (In Chapter Five we discuss additional variables that determine the use of information in constructing scale anchors and standards of comparison.)

The impact of scale anchoring discussed so far is referred to as the *perspective effect* in Ostrom and Upshaw's (1968) perspective theory and as the *range effect* in Parducci's (1982, 1983) range-frequency theory. Both theories maintain that respondents anchor the scale in a way that accommodates the range of stimuli presented to them. In addition, respondents try to discriminate between stimuli as best they can given the limits of the rating scale. Suppose, for example, that additional small squares, let us say A' to A'' , were inserted between squares A and B in the sequence of A to D squares shown in Figure 4.1. If respondents used the rating scale as shown in line 1, all four squares A to A'' would be assigned to the same rating category, resulting in limited discrimination. To be most informative, however, respondents would be unlikely to assign four

stimuli to one category of the response scale and one stimulus each to each of the remaining categories. Rather, they would most likely attempt to assign a roughly equal number of stimuli to each rating category, because "assigning the same number of stimuli to each category maximizes the number of bits of information transmitted (in the information theory sense)" (Parducci, 1983, p. 269). Parducci calls this the *frequency effect* in his range-frequency theory, and it is not addressed by Ostrom and Upshaw's model.

In sum, the use of rating scales is a function not only of the extreme stimuli that are used to anchor the scale (resulting in a range effect) but also of the number and distribution of stimuli (resulting in a frequency effect). Accordingly, a given stimulus will receive different ratings depending on the extremity, frequency, and distribution of context stimuli introduced in the questionnaire. Daamen and de Bie (1992) provide a detailed discussion of these issues and explore their implications for survey research.

Effect of Rank-Ordering Scales on Subsequent Ratings

A special complication may arise at the formatting stage when a rating task is preceded by a rank-ordering task: the direction of rank-ordering may affect the subsequent use of numerical rating scales. In a series of studies, Schwarz and Wyer (1985) asked respondents to rank-order a set of environmental issues from most important to least important or from least important to most important. Subsequently, respondents had to evaluate along a rating scale other environmental issues that were not included in the rank-ordering task. When respondents rated the importance of the issues, they rated all issues as more important if the earlier rank-ordering had progressed from most to least important than from least to most important. However, when they rated triviality of issues, they also rated all issues as more trivial when the preceding ranking had progressed from most to least important than least to most important. Moreover, the same pattern was obtained when the rankings pertained to triviality, as shown in Table 4.3.

TABLE 4.3 Mean ratings of environmental issues as a function of ranking criterion.

Rating of new items	Previous ranking of other issues by importance or triviality			
	Importance		Triviality	
	Most to least	Least to most	Most to least	Least to most
Importance	65.5	52.2	65.5	49.6
Triviality	47.4	42.6	49.5	29.5

Source: Adapted from Schwarz and Wyer (1985), Experiment 2.

Note: The possible range of ratings is from 0 ("not at all important" or "not at all trivial," respectively) to 100 ("extremely important" or "extremely trivial," respectively). $N = 18$ to 20 per condition.

In essence, respondents assigned higher ratings following the most-to-least sequence, independent of both the substantive implications of these ratings (importance or triviality) or the content of the rank-ordering task. Indeed, ranking desirable attributes in a spouse from most to least important also resulted in higher ratings of the importance and the triviality of environmental issues than did ranking desirable spousal attributes from least to most important.

That the impact of the direction of rank-ordering on the obtained ratings was unrelated to the content of the ranking task or to the substantive meaning of the ratings suggests that the observed effect was not driven by any substantive information that the rank-orderings may have brought to mind. Rather, the direction of rank-ordering influenced respondents' subsequent use of a rating scale. Apparently, respondents who first had to rank-order unrelated stimuli from most to least important (trivial, or whatever) subsequently focused on the end of the rating scale that reflected a high standing on the rating dimension, essentially the end equivalent to "most." In contrast, respondents who had to rank-order stimuli from least to most important (or trivial) subsequently focused on the low end of the rating scale, essentially the end

equivalent to "least." As a result, the obtained ratings were displaced toward the end of the scale that the respondents focused on, resulting in higher ratings when the ranking task elicited a focus on the high end and in lower ratings when it elicited a focus on the low end. Consistent with this assumption about the use of numeric rating scales, additional studies showed that the direction of preceding rankings did not affect evaluations along verbal response scales.

Context Effects at the Editing Stage

Finally, preceding questions may draw respondents' attention to issues of social desirability and self-presentation, resulting in biases in responses to subsequent questions to which these concerns may be relevant. This seems particularly likely to occur when the issue is not sufficiently sensitive to elicit these concerns spontaneously. We would expect different reports of tranquilizer consumption, for example, when the context questions pertain to medical issues or to issues of drug abuse. An amusing example of such effects was reported by Sherman and Golkin (1980). They told subjects about a father and son who were involved in a car accident. The father died on the spot but the son was rushed to the hospital and prepared for surgery. The surgeon came in, saw the patient, and exclaimed, "I can't operate on him; he's my son." This scenario seemed impossible to most people until they realized that the surgeon was the patient's mother, which many did only after the explanation was offered by the experimenter. However, following this involuntary confrontation with their own sexist biases, respondents went on to express particularly liberal attitudes on questions pertaining to gender-based discrimination.

Summary

The context in which a question is asked can influence respondents' cognitive processes at each stage of the question answering process. Moreover, the various processes are not mutually exclusive;

a given context effect may reflect the simultaneous operation of several processes. Finally, it is important to keep in mind that in some modes of administration context effects are not necessarily limited to the impact of preceding questions on subsequent ones. Under self-administered questionnaire conditions, for example, where respondents may read ahead or return to previous questions, questions asked later on may affect responses to preceding ones (see Schwarz & Hippler, 1995).

In the next chapter, we offer a more detailed exploration of context effects at the judgment stage. We will pay particular attention to the conditions under which information that is brought to mind by preceding questions results in assimilation or contrast effects in subsequent responses.